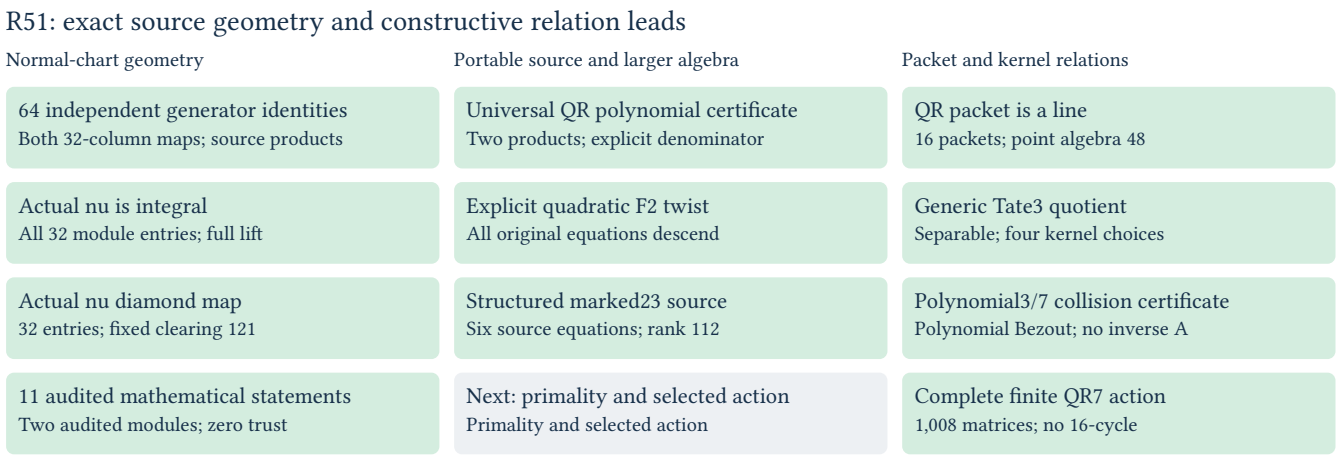
R51: integral coordinates and the actual QR7 packet

The ordinary universal QR variables are the slope and intercept of the line through P,2P,4P. The actual descended level 77 nu is integral in the retained normal-chart algebra and has a checked diamond formula. A full polynomial 3/7 kernel certificate gives controlled Velu collision support. The marked23 source now has a complete structured algebra of dimension 112 over F2(x).



Strict native receipts: results.json. Premises and proof scope: CM.md / PROOF_MAP.md.

Source coordinates and packet geometry

All 32 normalization generators have localized maps in both directions. A fresh process verifies 64 direct numerator products,both complete 32-column PID products and all 4 source-equation lifts. The actual nu has a full 32-entry PID lift. Its diamond/sign products and localized 32-entry lift pass,with fixed exponent 121=62+59 from the actual source coefficients.

For A=1+c,the QR packet line y=lambda*x+nu meets the Tate curve in `x³+(b+lambda²+A*lambda)x²+(A*nu+b*lambda)x+nu²+b*nu` The degree 48 point algebra has a separable cubic,source-compatible doubling, three-doubling identity and a full chord-recovery unit. A monic degree 24 point equation has a full source membership certificate. All 5 serialized products recheck. This supplies the actual packet coordinates for the next isogeny map.

Universal descent and larger source

The archived universal QR matrices clear to polynomial identities with D=(1+c)^5*Delta^3. Every original QR and triangular equation descends under the quadratic gamma twist;G_twisted=G+(1+c)^16. The marked23 source retains all 6 coupled equations and both complete matrices in auxiliary B,C. Actual base units,raw-to-normal f23-multiple identities and nu normalization pass. Its dimension 112 is an algebra statement;primality and selected 991 action are subsequent tests. CM.md states the full geometric and source premises.

Complete native record and formal scope

There are20 terminal attempts:16 accepted and4 rejected. Late markers after question-mark errors are rejected even with native exit 0. The expanded coefficient attempt hit its original 1200 s cap. No accepted producer was repeated to repair publication. Both compiled Lean artifacts are retained.

Attempt	Status	Receipt
base23-51-01	rejected	exit 124
base23-struct51-01	rejected	exit 0; 2 error(s)
base23-struct51-02	rejected	exit 0; 1 error(s)
base23-struct51-03	accepted	exit 0
collision51-01	accepted	exit 0
diamond-nu51-01	accepted	exit 0
diamond51-01	accepted	exit 0
fiber51-01	accepted	exit 0
glue-forward51-01	accepted	exit 0
glue-prep51-01	accepted	exit 0
glue-reload51-01	accepted	exit 0
glue-reverse51-01	accepted	exit 0
hecke3-51-01	accepted	exit 0
line51-01	accepted	exit 0
nu51-01	accepted	exit 0
overlap-lean51-01	accepted	compile 0; audit 0
specialize-lean51-01	accepted	compile 0; audit 0
twist51-01	rejected	exit 0; 2 error(s)
twist51-02	accepted	exit 0
universal51-01	accepted	exit 0

Eleven audited mathematical statements

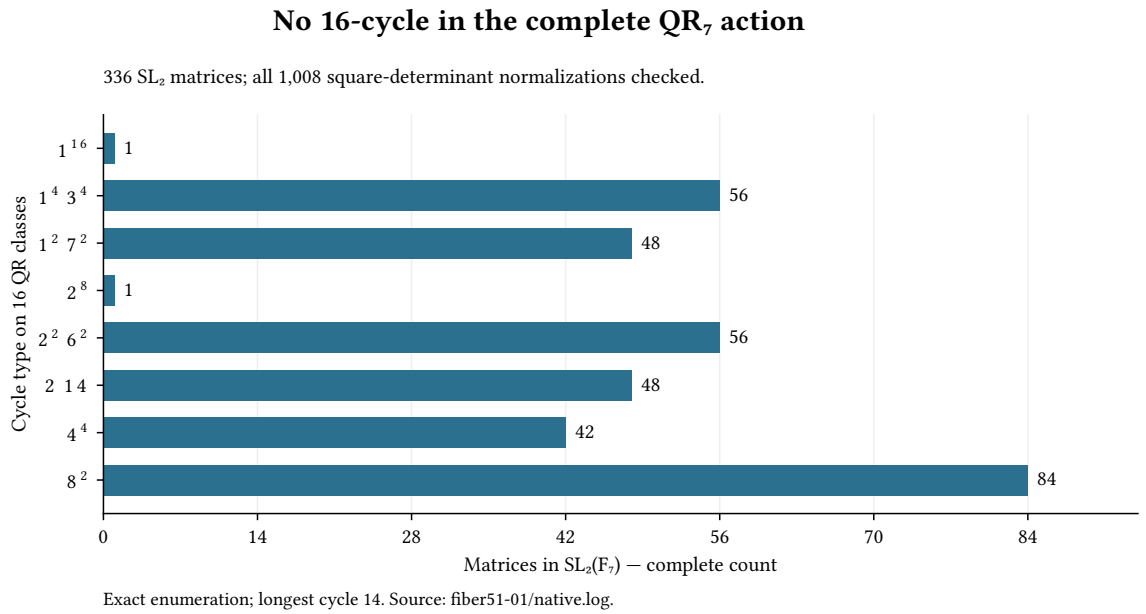
Overlap51 has 5 theorems:three exact characteristic2 overlap identities,a localized inverse,and inverse induced maps under injective compatible embeddings. Specialize51 has 6:denominator-unit cancellation,scaled ideal equality,the8P=P to7P=0 consequence,line intersection,quartic difference, and the cross-kernel inverse from the Bezout/point identities.

Both modules compile and pass the axiom audit with zero failing and zero trusted/native_decide declarations. Lean4.34.1/mathlibd13f23b remain pinned. Native matrices,actual point/Bezout identities,complete RR and geometric open-cover premises remain external inputs. Proper polarization and selected modular-factor action are separate constructions. See PROOF_MAP.md.

Decisive next normalization test

Construct multiplication by the now-integral nu on the full 32-generator algebra. Starting from the original source power-basis module,close under that matrix with the fixed Cayley-Hamilton bound 31. A full recovery of the normal basis would give the specific four-coordinate affine model;otherwise record the exact remaining PID cokernel/support. The old P67 RR embedding is retained. Then transport the cubic packet through Velu and verify the actual QR source equations,F2 descent and target action.

Finite action and direct-relation construction



The enumeration is complete,not sampled. It rules out a 16-cycle in the stated QR action;it does not obstruct generic connectedness. The first good odd split candidates are23 at level 77 and 11 at level 161. Their actual residual eigenvalues still need checking.

Direct relations: use T3 as an annihilator constraint

Generic Tate degree-3 quotient control
accepted

↓

Transport marked point and QR packet
not yet constructed

↓

Actual T3 action on retained Jacobian module
not yet computed

↓

Integral zero-eigenspace projector / scalar d
proposed Bezout or Smith certificate

↓

Selected target action and divisor saturation
requires actual source-bound certificates

CM prime 3 is inert; this is not an inferred nonzero CM map. See NEW-LEADS51.md.

The generic Tate3 map has full curve/order/unit products and serialized replay. The collision matrices give exact norms Delta,Delta^5,Delta^16 and U*psi3+V*psi7=Delta^16 with polynomial coefficients. No inverse A is used,so this certificate includes good A=0 fibers. The explicit cross-kernel inverse is Q*U(x)/Delta^16,using the quartic divided difference Q.

T3 is an inert-prime annihilator constraint. Construct its actual carrier correspondence and compatible integral or odd-adic module before claiming a projector scalar. A matrix only on F2 differentials does not certify that integral/sign decomposition. Primary sources and the external formalization lead are recorded in NEW-LEADS51.md;no external theorem is counted as locally audited. Raw custody and source-byte checks are separate from scientific replay.